Discussion week1

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Unit Conversion

1. *Convert the following units to SI units (some conversions are on the front cover of your book). Keep the correct number of significant figures.*

a. 21 feet
b. 1.4 feet/s
c. 3.87 feet/hour^2

Unit Conversion

$$21ft \times \frac{0.3048m}{1ft} = 6.4m$$

$$1.4ft/s \times \frac{0.3048m}{1ft} = 0.43m/s$$

$$\frac{3.87ft}{h^2} = \frac{3.87ft \times \frac{0.3048m}{1ft}}{(1h \times \frac{3600s}{h})^2} = 9.10 \times 10^{-8} m/s^2$$

Motion Diagram

2. Below is a motion diagram of somebody running a 100 meter dash.

a. *Draw position vectors at each time interval.*

b. *Draw the velocity vectors at each time interval.*

c. *Draw the acceleration vectors at each time interval.*

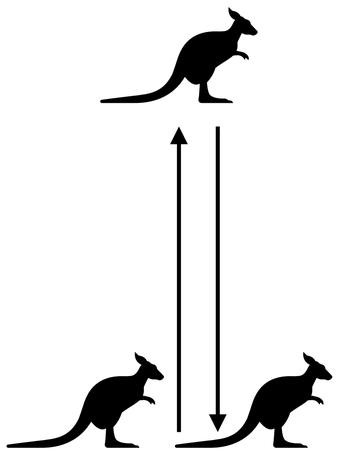
d. *If you were to define a coordinate system, where would be the most logical location for the origin and direction of the axes?*

e. *Plot the position vs. time graph for this runner.*

f. *Plot the velocity vs. time graph for this runner.*

Free Fall Motion

3. A kangaroo can jump straight up 3.2 m (according to one claim).

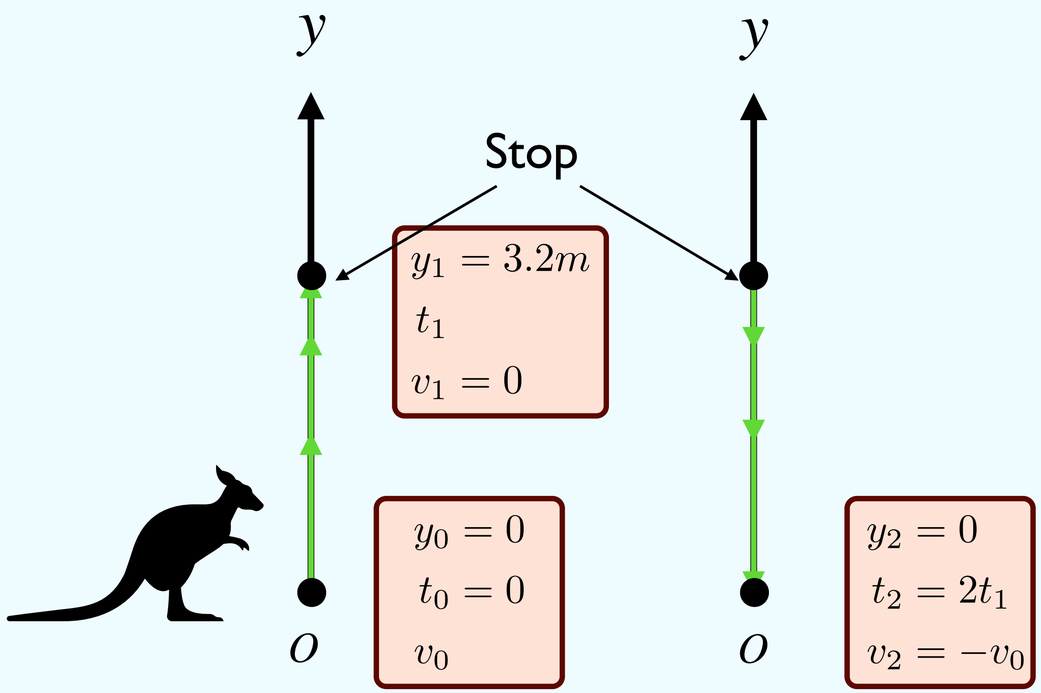


a. At what speed did the kangaroo leave the ground?

b. For how long was the kangaroo in the air?

Solving Problems in Physics

- Understand the problem and set up a model
  - Describe/explain in your own words
- Visualize
  - Draw the diagrams
  - Label relevant physical quantities (known and unknown)
- Identify the physics concepts
- Find the relationship between known and unknown quantities
  - Derive the symbolic expressions (Calculate numbers at the last step)
  - Check units
- Review the results
  - Reasonable values?
  - Correct units?



Displace the upward and downward motions

Known

$y_0 = 0$

$t_0 = 0$

$y_1 = 3.2m$

$v_1 = 0$

Find

$v_0 = ?$

$t_2 = ?$

$$a)\ v_0 = \sqrt{2gy_1}$$

$$b)\ t_2 = 2\sqrt{\frac{2y_1}{g}}$$

Visualization → *Physics concepts (Free-fall motion)* → *Equations*

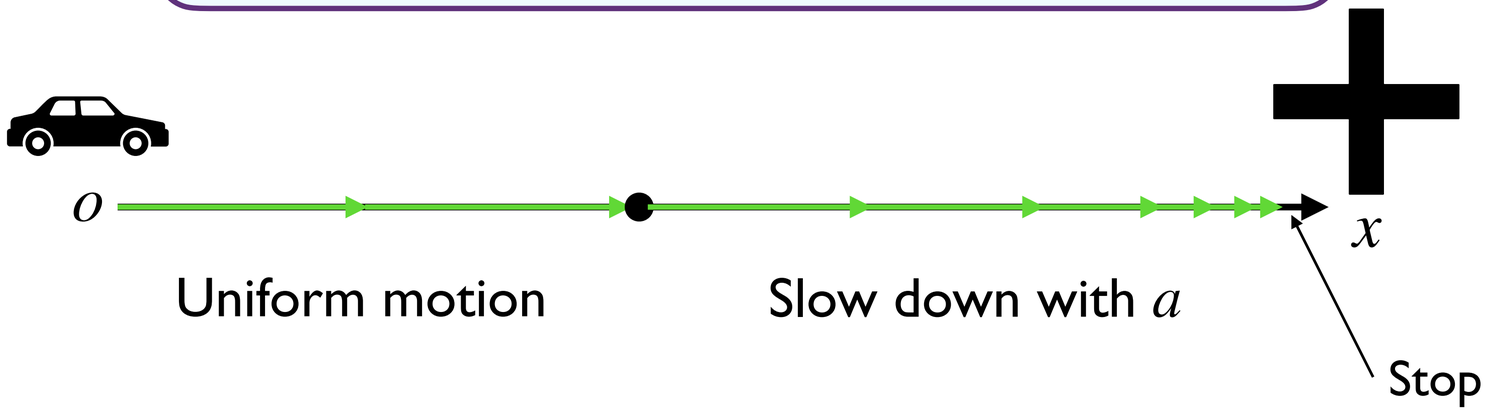
Unknown ⟷ *Known*

Car Braking Problem

You are driving to the grocery store at 20 m/s. You are 110 m from an intersection when the light turns red. Assume that your reaction time is 0.5 s and that your car brakes with constant acceleration.

a. How far are you from the intersection when you begin to brake?
b. What acceleration will bring you to a rest right at the intersection?
c. How long does it take you to stop after the light turns red?

Car Braking Problem



$$x_0 = 0$$
$$t_0 = 0$$
$$v_0 = 20m/s$$

$$x_1 = ?$$
$$t_1 = 0.5s$$
$$v_1 = 20m/s$$

$$x_2 = 110m$$
$$t_2 = ?$$
$$v_2 = 0$$

$$a) \Delta x = v_0 \times (t_1 - t_0) = 10m$$

$$b) v_2^2 - v_1^2 = 2a(x_2 - x_1) \rightarrow a = -2m/s^2$$

$$c) \frac{1}{2}|a|(t_2 - t_1)^2 = x_2 - x \rightarrow t_2 = 10s + 0.5s = 10.5s$$